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🌱 木

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Definition. 木 (*mù*, "tree; wood") is a **pictographic character** (象形字, *xiàngxíngzì*) and Kangxi radical 75. Its oracle-bone form depicts a standing tree: a vertical stroke (丅, trunk) bisected by a horizontal stroke (branches above, roots below). Formally, 木 is classified as a **primitive radical** — a graphemic atom that carries independent semantic content and composes productively with other elements to generate derived characters. As a left-hand component it is written 木 (4 strokes: 横 → 竖 → 撇 → 捺); stroke count and order are invariant across all derived forms.

Worked Example — Compositional Derivation. Consider the derivation chain:

$$木\,\mathrm{xrightarrow}\,+木林\,\mathrm{xrightarrow}\,+木森$$

- 林 (*lín*, "grove") doubles 木, encoding multiplicity $n = 2$.
- 森 (*sēn*, "dense forest") triples 木, encoding $n = 3$.

The spatial arrangement also carries meaning. Placing 木 with a horizontal mark **below** the midpoint yields 本 (*běn*, "root; origin"), while a mark **above** yields 末 (*mò*, "tip; end"). Letting $y = 0$ denote the trunk midpoint, the mark positions satisfy $y_{本} < 0 < y_{末}$: a topological encoding of position along the vertical axis that directly maps to the semantic opposition *origin* vs. *periphery*.

Key Theorem (Radical Compositionality). Let $\mathcal{R}$ be the set of Kangxi radicals and $\mathcal{C}$ the set of Chinese characters. Every character $c \in \mathcal{C}$ admits a unique **radical decomposition** $c = r \oplus s$, where $r \in \mathcal{R}$ is the classifying radical and s is a phonetic or semantic complement. For 木-family characters, $r =$ 木 and s shifts the semantic field within the domain of trees, wood, and plant material (e.g., 桌 *table*, 梯 *ladder*, 橙 *orange tree*). The radical therefore acts as a **type classifier**: knowing $r =$ 木 constrains the denotation to the arboreal semantic domain before the complement s is read.

Lab Cell.

```
from sympy import symbols, FiniteSet

# Represent 木-family characters as (radical, complement) pairs
radical = "木"
family = {
    "林": (radical, "木"),
    "森": (radical, "林"),
    "本": (radical, "mark_below"),
    "末": (radical, "mark_above"),
    "桌": (radical, "卓"),
}

print(f"Radical: {radical}")
print(f"Family size: {len(family)}")
for char, (r, s) in family.items():
    print(f"  {char} = {r} ⊕ {s}")
```

Running this cell confirms that 木 functions as a stable type label across all five derived characters, illustrating the compositionality theorem concretely.

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Definition (Character Composition). The logograph 林 (*lín*) denotes a *grove* or *forest* and belongs to the class of **ideographic compounds** (会意字, *huìyì zì*). Let $\mathcal{C}$ denote the set of Chinese characters and let $\oplus$ denote the horizontal juxtaposition operator on $\mathcal{C}$. Then

$$\text{林} = \text{木} \oplus \text{木},$$

where 木 (*mù*, "tree") is the base radical with stroke count $s(\text{木}) = 4$. The juxtaposition operator satisfies the **stroke-count additivity rule**:

$$s(\alpha \oplus \beta) = s(\alpha) + s(\beta), \qquad \alpha, \beta \in \mathcal{C}.$$

Hence $s(\text{林}) = 4 + 4 = 8$. Semantically, placing two identical elements side by side signals *collective density* in the domain denoted by the base element — a phenomenon termed the **Reduplication Principle**.

Worked Example. Decompose 林 into its component tree:

$$\text{林} \longrightarrow \underbrace{\text{木}}_{\text{left}} \Big| \underbrace{\text{木}}_{\text{right}} \longrightarrow \{\text{横}, \text{竖}, \text{撇}, \text{捺}\} \times 2.$$

Each 木 resolves into four primitive strokes: one horizontal (横), one vertical (竖), one left-falling (撇), and one right-falling (捺). The compositional depth of 林 is therefore $d = 2$, meaning two layers separate 林 from its irreducible strokes. In the HSK proficiency framework, 木 appears at Level 1 and 林 at Level 3, consistent with the depth increase.

Theorem (Semantic Chain by Reduplication). *Let $\alpha \in \mathcal{C}$ denote an atomic object of type T. Define $\alpha^{(k)}$ as the k-fold horizontal reduplication written as a single recognized character. Then the sequence $\alpha^{(1)}, \alpha^{(2)}, \alpha^{(3)}$ forms a monotone chain ordered by collective intensity.*

Corollary. For $\alpha = \text{木}$: 木 xrightarrow $k = 2$林 xrightarrow $k = 3$森 (*sēn*, dense forest), with stroke counts $4, 8, 12$ — an arithmetic progression with common difference $s(\text{木}) = 4$.

Lab Cell (SymPy).

```
from sympy import symbols, Eq, solve, Integer

s_mu = Integer(4)                       # 木 has 4 strokes
n    = symbols('n', positive=True, integer=True)

stroke_count = s_mu * n                 # strokes for n-fold reduplication of 木

for k, name in [(2, '林'), (3, '森')]:
    print(f"s({name}) = {stroke_count.subs(n, k)}")   # 8, 12

# Inverse query: which k gives a target stroke count?
target   = Integer(8)
solution = solve(Eq(stroke_count, target), n)
print(f"n = {solution} → 林")           # [2]
```

This arithmetic model formalizes the compositional transparency of Chinese ideographic compounds, linking visual structure, semantic rank, and numeric regularity within a single framework.

Payoff

Think about the moment you first learned that two words could combine into one new idea — like "rain" and "coat" becoming "raincoat." Something clicks: language builds on itself. That same click is waiting for you inside 林.

Picture this. You already know 木 (*mù*) — one tree, standing alone. Now place a second 木 right beside it. You get 林 (*lín*): a grove, a forest, a gathering of trees. The meaning did not just add; it multiplied. One tree is a fact. Two trees together become a *place*.

The simple rule. In Chinese character logic, doubling a root signals plurality or intensification:

$$\text{木} + \text{木} = \text{林} \quad \text{(grove)}$$

And the pattern keeps going — triple 木 gives you 森 (*sēn*), a dense, ancient forest. A single symbol carries an entire ecosystem of meaning, compressed into strokes you can write in seconds.

Why this is the natural endpoint. Every character you have studied so far was a seed. 林 shows you what happens when seeds grow together — new meaning emerges from combination, not just accumulation. This is the engine behind thousands of Chinese words.

Try it yourself. Find one other doubled-root character (hint: look up 日 — *sun*). What does doubling it create? Then, if you are curious, dive into how 林 appears inside compound words like 森林 (*sēnlín*, jungle) — a small adventure that opens the whole forest.